Algo HW-1

Q.1. $a = qb + r$

$a = -33$, $b = 7$, $r = 2$

$\therefore -33 = 7q + 2$

$\therefore 7q = -35$

$\therefore q = -5$

$0 \leq r < b$

$\therefore 0 \leq 2 < 7$

$\because q = -5$ is an integer and r lies between 0 and 7, The remainder of -33 and 7 is 2

Q.2 $x | y$

Let there be an integer q_1

$\therefore y = q_1 x$

$y | z$

Let there be an integer q_2

$\therefore z = q_2 y$

$\because y = q_1 x$

$\therefore z = q_2 (q_1 x)$

$\therefore z = (q_1 q_2) x$

$\therefore x | z$

$\because$ Since q_1 and q_2 are integers, By formal definition of divisibility, $x | z$

Q.3. $a_1 = bq_1 + r_1$

$a_2 = bq_2 + r_2$

$a_1 a_2 - r_1 r_2 = (bq_1 + r_1)(bq_2 + r_2) - r_1 r_2$

$a_1 a_2 - r_1 r_2 = b^2 q_1 q_2 + bq_1 r_2 + bq_2 r_1 + r_1 r_2 - r_1 r_2$

$\therefore a_1 a_2 - r_1 r_2 = b^2 q_1 q_2 + bq_1 r_2 + bq_2 r_1$

$\therefore a_1 a_2 - r_1 r_2 = b(bq_1 q_2 + q_1 r_2 + q_2 r_1)$

Since b, q_1, q_2, r_1 and r_2 are integers,

$\therefore b \mid a_1 a_2 - r_1 r_2$

Q.4. $f_1(n) = O(f_2(n))$, $f_1(n) > f_2(n)$ for all $n > 0$

$\therefore f_1(n) \leq c \cdot f_2(n)$ for all $n \geq n_0$

Lets take $c = 2$, and $n_0 = 1$

$\therefore$ There exists positive integers of c and n_0

Q.5. $f_2(n) = O(f_1(n))$

$\therefore f_2(n) \leq c \cdot f_1(n)$ for all $n \geq n_0$

If we take $c = 2$ and $n_0 = 1$ then there exists positive integers of c and n_0

$f_1(n) + f_2(n) = O(f_1(n))$

Let there be integers, c_2 and n_2

$\therefore f_1(n) + f_2(n) \leq c_2 \cdot f_1(n)$ for all $n \geq n_0$

If we take $c = 3$ and $n_0 = 2$ there exists positive integers of c and n_0.

Q.6. Let i be an integer such that $i \geq 2$

$\therefore f_i(n) = O(f_i(n))$

Let c_i and n_i be positive integers.

Let k be an integer.

$\therefore f_1(n) + f_2(n) + \dots + f_k(n) = O(f_1(n))$ for all $n \geq n_0$

$\therefore f_1(n) + f_2(n) + \dots + f_k(n) \leq (c_1 + c_2 + \dots + c_k) f_1(n)$

$f_1(n) + f_2(n) + \dots + f_{k+1}(n) = (c_1 + c_2 + \dots c_k) f_1(n) + (c_{k+1}) f_1(n)$

$\therefore f_1(n) + f_2(n) + \dots + f_{k+1}(n) = (c_1 + c_2 + \dots + c_{k+1}) f_1(n)$

As $(c_1 + c_2 + \cdots + c_{k+1})$ is an integer ~~and hence~~ ~~greate~~ Hence, proved.